

Experiment No: 1

Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required:

JavaScript, HTML, CSS and Code Editor.

Theory:

JavaScript is a lightweight, interpreted, and object-oriented programming language used to make web pages interactive and dynamic. It is one of the core technologies of web development, along with HTML and CSS. JavaScript can manipulate webpage content, respond to user actions, validate forms, display messages, and interact with web browsers.

JavaScript can be included in a webpage in three different ways:

1. Inline JavaScript

Inline JavaScript is written directly within an HTML element using event attributes such as onclick, onmouseover, etc. It is suitable for simple tasks but is not recommended for large applications because it mixes HTML and JavaScript code.

2. Internal JavaScript

Internal JavaScript is written inside the <script> tag within the HTML document. It is useful when the script is required only for a specific webpage.

3. External JavaScript

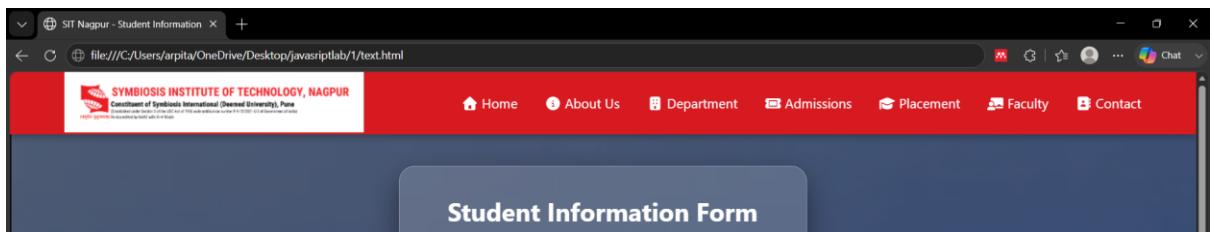
External JavaScript is stored in a separate .js file and linked to the HTML document using the src attribute of the <script> tag. This approach improves code organization, reusability, and maintainability.

Console Methods in JavaScript

The browser console is used for debugging and testing JavaScript code. Common console methods include:

- **console.log()** – Displays general information or messages.
- **console.error()** – Displays error messages.
- **console.warn()** – Displays warning messages.
- **console.info()** – Displays informational messages.
- **console.table()** – Displays data in tabular form.

Output



Case Study Program Code

Problem Statement

Design a student information webpage that accepts details such as Name, Roll Number, Age, Email, Phone Number, Address, Department, Semester, CGPA, Gender, Professional Skills, and Technical Skills. After clicking the **Submit** button, display a welcome message and show all entered details dynamically on the webpage.

Implementation

The webpage consists of an HTML form where the user enters student details. JavaScript is used to retrieve the entered data using **DOM methods**, display a welcome message through **alert()**, and generate the student information dynamically using **innerHTML**. The Submit button calls the JavaScript function through an **inline event (onclick)**, while the function itself is written using **internal JavaScript**.

Conclusion:

JavaScript allows developers to add interactivity and dynamic behaviour to web pages. By using inline, internal, and external JavaScript, along with various console methods, user information and welcome messages can be displayed effectively, enhancing the overall user experience.